

Nathan Potraz

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EXPERIENCE

Hardware & Software Engineer (Founder)

Self-Employed • June 2022 – Present • Oceanside, CA

- Engineered **over 100 custom-tuned** competitive gaming controllers, performing precise circuit-board soldering and firmware optimization to meet professional-grade performance standards.
- **Consulted directly with clients** to define technical requirements, translating user performance needs into custom hardware and software specifications.
- **Managed full-cycle product development**, including supply chain logistics, hardware debugging, and providing post-deployment technical support.
- **Contributed to an open-source motherboard project** by performing board testing, taking electrical measurements, and authoring technical documentation to improve community hardware standards.

PROJECTS

Devlog | devlog.ntpotraz.dev/ • Oct 2025 – Present

- A journaling web application built using **TypeScript React** and a **Golang HTTP server and API**.
- Features simple and fast journal entry **creation, sharing, and editing** utilizing **SQLite hosting** with Turso.
- Integrated Clerk for **secure user authorization**, supporting GitHub and Google SSO as well as email/password.
- Automated deployment with GitHub Actions and Docker, reducing production release time from 15 minutes to under 3 minutes via a CI/CD pipeline targeting Google Cloud Run.

PIC Health & Resource Mobile App | ntpotraz.github.io/PIC-Health-App/ • Aug 2024 – Dec 2024

- Developed a resource and event calendar **React Native mobile application** for underrepresented communities.
- **Led a team of 5 developers** to work alongside our sponsor to **plan, design, and develop** our Capstone project.
- Set up GitHub repo, Expo project, and assisted my dev team with their environments and using version control.
- Integrated **Google API** to fetch and render **community events** for users from the admin Google Calendar.

Japanese-English Translator | github.com/nthpotraz/cs421-translator/ • Mar 2024 – May 2024

- **Engineered a C++ natural language processor** using a **Deterministic Finite Automata Scanner** to tokenize and categorize complex linguistic inputs with 100% lexical accuracy.
- Built a Recursive Descent Parser to enforce strict grammar rules, implementing a look-ahead mechanism to handle multi-token syntax structures efficiently.
- Optimized **dictionary lookups to O(1) average time complexity** by implementing custom **Hash Maps**, enabling near-instantaneous translation of large text files compared to standard linear search methods.
- Developed a **translation engine** that programmatically reordered tokens to account for syntactic differences between Japanese and English, ensuring grammatically coherent output.

Gloomy Monsters | github.com/ntpotraz/gloomy-monsters/ • Mar 2020 – Mar 2020

- Created a React-based companion app to **manage complex board game mechanics** and stat tracking for users.
- Designed a **nested state architecture to handle hierarchical data** for multiple active monster groups.
- Integrated a **complex JSON dataset to dynamically populate unit statistics** and ability deck behaviors.
- Applied **conditional visual styling to adjust UI elements** based on specific game rules and unit types.

EDUCATION

B.S in Computer Science

California State San Marcos • San Marcos, CA • 3.7 GPA • Cum Laude Honors

Relevant Coursework: OOP, Data Structures & Algorithms, Networking, Cloud, Web Programming, Databases, Computer Security, AI, Operating Systems, AGILE/SCRUM.

SKILLS

Backend: Golang, Python, Java, C++, SQL, SQLite, MySQL, PostgreSQL

Frontend: TypeScript, JavaScript, PHP, React, React Native

DevOps/Tools: Linux, Docker, CI/CD, Git, Agile, AWS, GCP